Deep learning analysis of nanoalloys from HRTEM images

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Introduction

Electron microscopy allows us to study the structure of objects much smaller than the wavelength of visible light. In the field of nanoparticles, high-resolution transmission electron microscopy (HRTEM) enables us to investigate the structure and geometry of nanoalloys. However, analyzing HRTEM images can be time-consuming and challenging. It needs a lot of expertise and experience in the imaging field to analyze them correctly. The use of deep learning can be an option to accelerate and simplify the analysis of these HRTEM images and exctract information about the structure and properties of nanoalloys. But for this, a dataset is needed to train the deep learning models. This dataset is generated using a molecular dynamics code developed in the laboratory.

To start this PhD project, we focus on the **Ag–Co** system, an immiscible alloy where the structure is known. In this system, **Ag** atoms preferentially segregate to the surface while **Co** atoms tend to form the core, consistent with their lower and higher surface energies, respectively. The objective with this system is to check if the deep learning models can learn and extract the same information from the HRTEM images. Once this objective is done, other systems with more complex structures that we don't know yet will be studied to extract new information about their structure and properties.

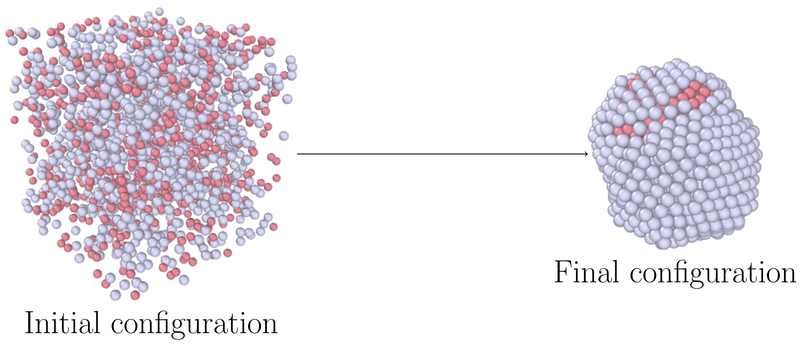
Molecular dynamic (MD) simulation

- **Number of atoms:** between 10 and 5000 atoms ;
- **Simulation time:** between 0.5 ns and 5 ns ;
- **Composition:** between 0% and 100% of **Ag** atoms (the rest being **Co** atoms) ;
- **Initial temperature:** between 1500 K and 2500 K ;
- **Final temperature:** 300 K ;
- **Canonical ensemble:** Nosé-Hoover thermostat + linear decrease of the temperature

$$T(t) = \frac{T_{\text{final}-T_{\text{initial}}}}{t_{\text{final}}} t + T_{\text{initial}} \tag{1}$$

- **Interatomic potential:** TBSMA (Tight-Binding Second Moment Approximation)

$$\mathcal{U} = \sum_{i=1}^{N} \left[\sum_{j \neq i}^{N} A e^{-p(r_{ij}/r_0 - 1)} - \sqrt{\sum_{j \neq i}^{N} \xi^2 e^{-2q(r_{ij}/r_0 - 1)}} \right] \tag{2}$$

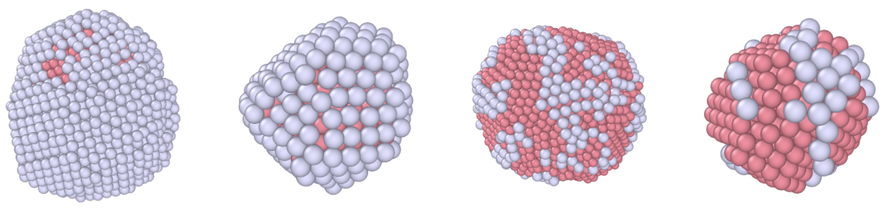


Creation of the dataset

Nanoalloys dataset

The **dataset of nanoalloys** is generated by using **MD simulations with random parameters**, as described in the "Molecular dynamic (MD) simulation" block.

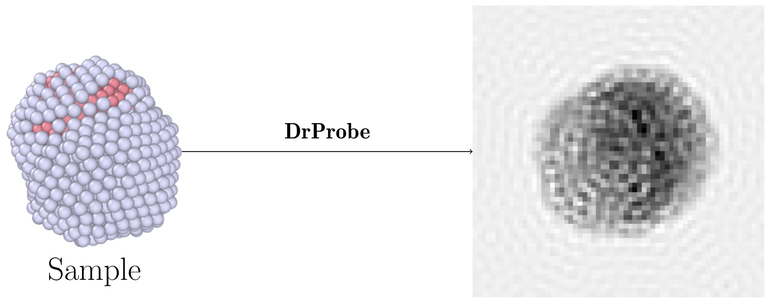
- **Number of data:** 48000 data points ;
- **Large diversity of data** thanks to the random parameters of the MD simulations.



Examples of the dataset

HRTEM images dataset

The **dataset of HRTEM images** is generated by using the **DrProbe** software to simulate the HRTEM images of nanoalloys in the nanoalloysdataset.



Neural network

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$$\eta = 2\left|\frac{n_{Ag}^{out}}{n_{Ag}^{out} + n_{Co}^{out}} - \frac{n_{Ag}}{n_{tot}}\right| + 2\left|\frac{n_{Ag}^{in}}{n_{Ag}^{in} + n_{Co}^{in}} - \frac{n_{Ag}}{n_{tot}}\right| - \frac{d_{com}}{2r_g} \tag{3}$$

Outlook

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References

[1] Cleri, 1993 [2] DrProbe